



Review Day
Sit
Anywhere

Continuity

Clarification and details

limits and continuity
A function f is continuous from the right at a number a if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and f is continuous from the left at a if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

A function f is **continuous on the open interval** (a, b) [or (a, ∞) or $(-\infty, b)$ or $(-\infty, \infty)$] if it is continuous at every number in the interval. It is **continuous on the closed interval** $[a, b]$ if it is continuous on (a, b) and is also continuous from the right at a and continuous from the left at b .

Similarly, one can define continuity of functions on other types of intervals such as $[a, b)$, $(a, b]$, $[a, \infty)$, and $(-\infty, b]$. For instance, f is continuous on $[a, b)$ if it is continuous on (a, b) and continuous from the right at a .

EXAMPLE 4

on (a, b) and continuous from the right at a .

EXAMPLE 6 Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

Solution If $-1 < a < 1$, then using the Limit Laws, we have

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) \\&= 1 - \lim_{x \rightarrow a} \sqrt{1 - x^2} && \text{(by Laws 2 and 7)} \\&= 1 - \sqrt{\lim_{x \rightarrow a} (1 - x^2)} && \text{(by 11)} \\&= 1 - \sqrt{1 - a^2} && \text{(by 2, 7, and 9)} \\&= f(a)\end{aligned}$$

Thus, by Definition 1.16, f is continuous at a if $-1 < a < 1$. We must also calculate the right-hand limit at -1 and the left-hand limit at 1 .

$$\begin{aligned}\lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} (1 - \sqrt{1 - x^2}) \\&= 1 - \sqrt{\lim_{x \rightarrow -1^+} (1 - x^2)} && \text{(as above)}\end{aligned}$$

$$\begin{aligned} &= 1 - \sqrt{1 - 1^2} \\ &= 1 = f(-1) \end{aligned}$$

So f is continuous from the right at -1 . Similarly,

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (1 - \sqrt{1 - x^2}) \\ &= 1 - \sqrt{\lim_{x \rightarrow 1^-} (1 - x^2)} = 1 - 0 = 1 = f(1) \end{aligned}$$

So f is continuous from the left at 1 . Therefore, according to Definition 1.18, f is continuous on $[-1, 1]$.